

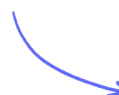
Calculator Questions

Algebraic Fractions

Simplifying Algebraic Fractions / Adding & Subtracting Algebraic Fractions /
Multiplying & Dividing Algebraic Fractions / Solving Equations with Algebraic
Fractions

Medium (7 questions)	/19
Hard (7 questions)	/27
Very Hard (5 questions)	/21
Total Marks	/67

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Medium Questions

1 Simplify.

$$\frac{3}{x-1} - \frac{2}{2x+1}$$

(3 marks)

2 Simplify.

$$\frac{5}{3x} \times \frac{9x}{20}$$

(2 marks)

3 Simplify.

$$\frac{x^2 + 5x}{x^2 - 25}$$

(3 marks)

4 Write as a single fraction in its simplest form.

$$\frac{x-5}{3} + \frac{6}{x+2}$$

(3 marks)

5 Simplify $\frac{x^2 - 25}{x^2 - 2x - 35}$.

(3 marks)

6 Write $\frac{x-8}{x} + \frac{3x}{x+1}$ as a single fraction in its simplest form.

(3 marks)

7 Simplify.

$$\frac{x^2 - 9}{x^2 - 2x - 3}$$

(2 marks)

Hard Questions

1 Simplify.

$$\frac{5p^2 - 20p}{2p^2 - 32}$$

(3 marks)

2 Simplify $\frac{x^2 - 25}{x^2 - 17x + 60}$.

(4 marks)

3 $f(x) = 7x - 4$ and $g(x) = \frac{2x}{x-3}$, $x \neq 3$

Find $\frac{f(x)}{2} + g(x)$.

Give your answer as a single fraction, in terms of x , in its simplest form.

(3 marks)

- 4 Write $\frac{x}{2} - \frac{2x+4}{x+1}$ as a single fraction, in its simplest form.

(3 marks)

- 5 Solve.

$$\frac{1}{x} - \frac{2}{x+1} = 3$$

Show all your working and give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(7 marks)

- 6 $f(x) = \frac{3}{x+2}$, $x \neq -2$ and $g(x) = 8x - 5$.

Write $g(x) - f(x)$ as a single fraction in its simplest form.

(3 marks)

7 Simplify.

$$\frac{2x^2 - x - 1}{2x^2 + x}$$

(4 marks)

Very Hard Questions

1 Simplify $\frac{2x^2 + x - 15}{ax + 3a - 2bx - 6b}$.

(5 marks)

- 2 A car completes a 200 km journey at an average speed of x km/h.
The car completes **the return journey** of 200 km at an average speed of $(x + 10)$ km/h.

i) Show that the difference between the time taken for each of the two journeys is

$$\frac{2000}{x(x + 10)} \text{ hours.}$$

[3]

ii) Find the difference between the time taken for each of the two journeys when $x = 80$.
Give your answer in **minutes** and **seconds**.

..... min s [3]

(6 marks)

3 $f(x) = 8 - 3x$ and $g(x) = \frac{10}{x+1}$, $x \neq -1$.

Write $f(x) + g(x)$ as a single fraction in its simplest form.

(3 marks)

4 $2 - \frac{x+2}{x-3} - \frac{x-6}{x+3}$ can be written as a single fraction in the form $\frac{ax+b}{x^2-9}$ where a and b are integers.

Work out the value of a and the value of b .

(4 marks)

5 Show that $\frac{1}{6x^2+7x-5} \div \frac{1}{4x^2-1}$ simplifies to $\frac{ax+b}{cx+d}$ where a , b , c and d are integers.

(3 marks)